Customizing Large Vision Model–Guided Low-Rank Approximation for Ground-Roll Denoise

Jiacheng Liao, *Graduate Student Member, IEEE,* Feng Qian, *Senior Member, IEEE,* Ziyin Fan, Yongjian Guo

***Abstract*—ground-roll is a dominant source of coherent noise in land and vertical seismic profiling (VSP) data, severely masking reflection events and degrading subsequent imaging and interpretation. Conventional attenuation methods, including transform-domain filtering, sparse representation, and deep learning, often suffer from limited adaptability, signal leakage, or dependence on labeled training data, especially under strong signal–noise overlap. To address these challenges, we propose a training-free framework that reformulates ground-roll attenuation as a semantic-guided signal separation problem. Specifically, a promptable large vision model is employed to extract high-level semantic priors by converting seismic gathers into visual representations and localizing ground-roll-dominant regions via text or image prompts. The resulting semantic response is transformed into a continuous soft mask, which is embedded into a mask-conditioned low-rank inverse formulation to enable spatially adaptive suppression and reflection-preserving reconstruction. The resulting semantic response is transformed into a continuous soft mask, which is embedded into a mask-conditioned low-rank inverse formulation to enable spatially adaptive suppression and reflection-preserving reconstruction. An efficient alternating direction method of multipliers (ADMM)-based solver is further developed to solve the proposed inverse problem, enabling stable and physically consistent signal recovery without requiring task-specific training or manual annotation. Extensive experiments on both synthetic and field VSP datasets demonstrate that the proposed method achieves superior ground-roll attenuation while preserving reflection continuity and waveform fidelity, consistently outperforming representative transform-domain filtering and implicit neural representation methods.**

***Index Terms*—ground-roll, low-rank optimization, large vision model, alternating direction method of multipliers (ADMM).**

I. Introduction

ground-roll is a type of surface wave generated by the seismic source that propagates in the near-surface layer, where it can be scattered by surface heterogeneities [1, 2]. This propagation and scattering behavior concentrates its energy near the surface, causing ground-roll to exhibit high amplitude, low frequency, and low apparent velocity [3, 4]. This significantly degrades the signal-to-noise ratio and imaging quality of seismic data, thereby hindering subsequent velocity analysis, imaging, and geological interpretation [1, 5]. Furthermore, ground-roll interferes with reflections in both the time–space and frequency domains: it typically forms a fan- or cone-shaped energy pattern in the time–space domain and exhibits a broadband, dispersive spectrum overlapping the reflection band in the frequency domain [5–8]. As a result, ground-roll and reflections are difficult to separate in the time–space domain [9].

To suppress ground-roll, sparse transform-based methods remain a mainstream choice in the seismic literature. These methods can be broadly grouped into three representative categories: Fourier-analysis-based approaches, time–frequency analysis, and dictionary-learning (DCL). Fourier-analysis-based approaches (e.g., F–K filtering [8, 10] and the Radon transform [11]) attenuate ground-roll by mapping seismic gathers into transform domains and exploiting the differences in apparent velocity and/or dip between ground-roll and reflections. Compared with Fourier-analysis-based methods, time–frequency analysis methods generally provide stronger sparse representation capability. They expand seismic data using multiscale time–frequency basis functions, such as wavelet [12] and curvelet bases [13], so that the energy of dispersive ground-roll becomes more concentrated in the time–frequency domain.

Compared with the first two families of methods that rely on fixed sparse domains, DCL methods [14] generally provide stronger sparse representation capability by learning a data-adaptive dictionary. Through sparse coding and reconstruction of seismic records, signal components can be represented by only a few nonzero coefficients, while noise is suppressed during sparse approximation. However, sparse transform-based methods methods still rely on the assumption that signal and noise are separable in the representation domain [15]. When ground-roll and reflections strongly overlap, the separation often becomes difficult..

Deep learning (DL) methods learn a mapping from contaminated gathers to clean reflections or noise components [16]; from the learning paradigm perspective, it can be broadly categorized into supervised and unsupervised/self-supervised approaches. Supervised models (e.g., U-Net, ResNet, GANs) can capture the coherent textures of ground-roll and provide fast inference [17, 18], but they are limited by scarce clean labels and weak cross-area generalization [16]. Recent trends therefore move toward physics-constrained and self-/unsupervised schemes, including label-free per-dataset adaptation such as generative models and implicit neural representations (INR) [4, 19–22]. However, these methods often require per-dataset optimization with carefully tuned objectives and hyperparameters [23], leading to limited robustness across datasets.

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To address the challenges of ground-roll suppression, we introduce a novel few-shot large-vision-model-guided low-rank approximation (LVM-LRA) framework, which couples two complementary modules with clear functional roles. The LVM provides promptable semantic/morphological recognition of ground-roll-dominant regions [24], whereas the LRA performs a physics-consistent low-rank decomposition to separate ground-roll from reflections [25]. This division of labor converts high-level recognition into an explicit spatial prior that guides the subsequent separation, leading to the following advantages:

1) *High denoising fidelity:* By converting prompt-driven semantic recognition into an explicit spatial prior, the proposed framework constrains the separation to ground-roll-dominant regions while enforcing low-rank modeling. This semantic guidance reduces ambiguity in strong overlap zones and promotes more faithful recovery of reflection continuity, leading to improved suppression accuracy with reduced signal leakage.
2) *Robustness and cross-dataset transfer:* The LVM component requires only a few prompts (including a lightweight few-shot exemplar) to adapt to different surveys, and the subsequent LRA stage involves a small set of physically interpretable parameters [26]. This combination reduces the tuning cost, enabling strong cross-dataset transferability.
3) *Computational efficiency:* The LVM produces the semantic mask with fast inference, and the LRA optimization has a clear computational profile dominated by a few low-rank updates, as shown in IV-C. Compared with DL methods requiring costly network training and large-scale inference, the overall pipeline is computationally lightweight and well-suited for large field datasets.
4) *Improved interpretability:* Although large vision models can be opaque, their behavior is explicitly controlled by human-readable prompts and visual exemplars, making the semantic intent transparent. The LRA stage further provides a physics-consistent and mathematically explicit decomposition, offering interpretable outputs (ground-roll and reflection components) and controllable processing behavior [27].

Building on these advantages, LVM-LRA emerges as a promising solution for removing ground-roll, which constitutes the central objective of this study. To further distinguish the approach from existing ground-roll-suppression methods [10, 17, 19], three key technical innovations are introduced to enhance the effectiveness of the LVM-LRA framework. The main contributions are summarized as follows:

The main contributions of this study are summarized as follows:

1) *LVM with hybrid prompting for ground-roll masking:* This study adopts hybrid prompting to drive dense prediction: morphological appearance descriptions, physics-informed propagation prompts, and a few-shot visual exemplar are jointly combined into a multimodal prompt set and fed into the LVM to produce a probabilistic response map of ground-roll-dominant regions, thereby generating the corresponding ground-roll mask. To the best of our knowledge, this work is the first to introduce LVM into ground-roll suppression.
2) *Mask-guided LRA model for ground-roll separation:* We formulate ground-roll suppression as a decomposition problem and impose low-rank regularization on both the reflection and ground-roll components to promote coherent, predictable structures while suppressing unstructured fluctuations. In addition, a mask is embedded into the LRA model as an explicit spatial support constraint, so that the ground-roll component is modeled and estimated only within the detected ground-roll-dominant region.
3) *Efficient ADMM-based optimization strategy:* For the proposed LVM-LRA decomposition model, an efficient optimization strategy based on the alternating direction method of multipliers (ADMM) is developed by splitting the overall objective into several interpretable subproblems that are optimized in an alternating manner [28].

Extensive experiments on both synthetic and field VSP datasets demonstrate that the proposed framework achieves robust ground-roll suppression while preserving reflection continuity and waveform integrity, consistently outperforming representative transform-domain filtering and learning-based baselines.

The remainder of this article is organized as follows. Section II presents the problem statement and semantic-guided inverse formulation. Section III introduces the semantic prior extraction stage using a promptable vision model. Section IV describes the mask-conditioned low-rank reconstruction and ADMM solver. Section V details the datasets, baselines, and evaluation protocol. Section VI presents quantitative and qualitative results, including leakage analysis and spectral validation. Finally, Section VII concludes the article.

II. Problem Statement and Formulation

We consider ground-roll attenuation as an additive signal decomposition problem. The observed seismic gather $\mathbf{Y} \in \mathbb{R}^{T\times X}$ is modeled as the superposition of reflection signals, ground-roll, and residual noise:

$$\mathbf{Y} = \mathbf{X} + \mathbf{G} + \mathbf{N}, \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{T\times X}$ denotes the reflection-dominant component, $\mathbf{G} \in \mathbb{R}^{T\times X}$ represents the ground-roll component, and $\mathbf{N}$ accounts for residual noise. Recovering $\mathbf{X}$ and $\mathbf{G}$ from $\mathbf{Y}$ is inherently ill-posed, as the decomposition is non-unique without additional prior information . A common approach is to introduce regularization and formulate the problem as

$$\min_{\mathbf{X},\mathbf{G}} \frac{1}{2}\|\mathbf{Y}-\mathbf{X}-\mathbf{G}\|_F^2 + \lambda_{\mathrm{s}} R(\mathbf{X}) + \lambda_{\mathrm{g}} R(\mathbf{G}), \tag{2}$$

where $R(\mathbf{X})$ and $R(\mathbf{G})$ encode prior assumptions on the reflection and ground-roll components, respectively.

However, due to the severe time–space overlap between reflections and ground-roll, together with the strongly non-stationary and dispersive nature of ground-roll, imposing a global constraint on $\mathbf{G}$ is physically unreasonable and often leads either to over-suppression of weak reflection signal

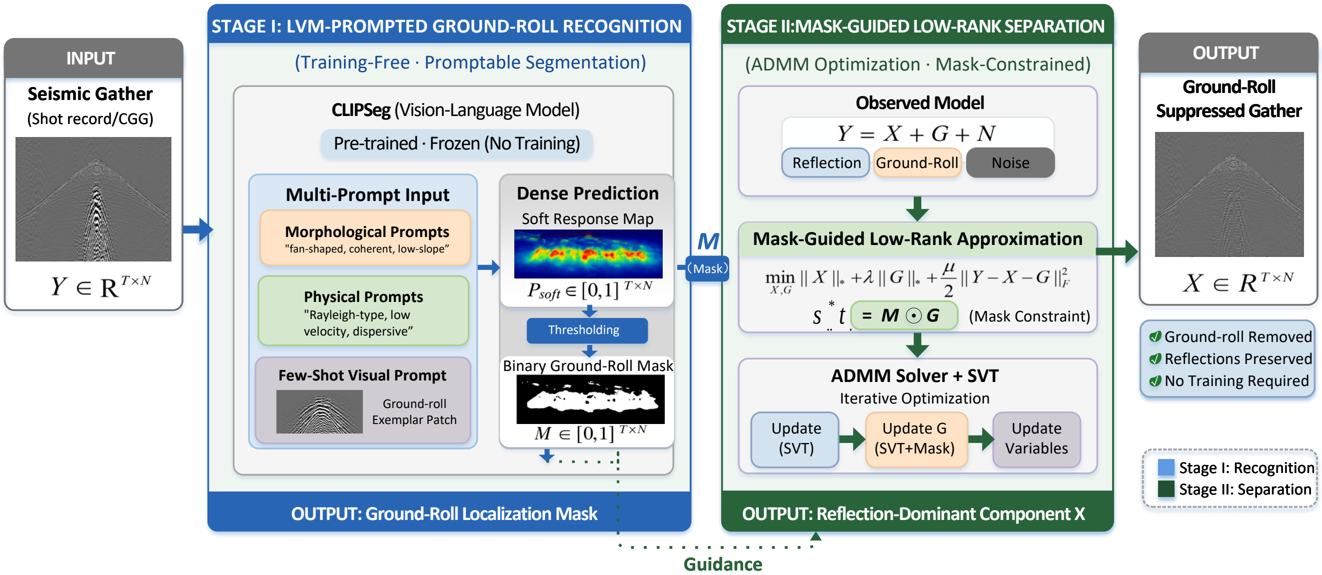


Fig. 1. Schematic diagram of the proposed LVM-LRA model.

or to noticeable residual ground-roll artifacts, particularly in overlapping regions.Motivated by [29–31], we introduce a binary spatial mask $\mathbf{M} \in \{0,1\}^{T\times X}$ to explicitly characterize the ground-roll-dominant support region. The ground-roll component is assumed to be spatially supported only within the masked region, which can be expressed as

$$\mathbf{G} = \mathbf{M} \circ \mathbf{G}, \tag{3}$$

where $\circ$ denotes the Hadamard product.

Rather than modifying the seismic data directly, the mask serves as a spatial prior that constrains the modeling and suppression of ground-roll. Existing approaches typically estimate such masks using heuristic hard thresholds [29] or task-specific deep learning models [30, 31]. In contrast, we propose a new LVM-guided mask generation strategy that leverages high-level semantic and structural information to localize ground-roll regions with a few-shot visual exemplar. Incorporating the mask into the optimization framework leads to the following mask-guided formulation:

$$\min_{\mathbf{X},\mathbf{G}} \frac{1}{2}\|\mathbf{Y} - \mathbf{X} - \mathbf{G}\|_F^2 + \lambda_s P(\mathbf{X}) + \lambda_g P(\mathbf{G}), \tag{4}$$

This formulation enables selective suppression of ground-roll within the detected noise-dominant region while preserving reflection energy elsewhere, thereby achieving a controllable balance between noise attenuation and signal preservation.

III. Stage I: CLIPSeg-Based Multimodal Prompting for Ground-Roll Masking

Stage I aims to generate a ground-roll mask for the subsequent masking-guided separation. This section first introduces the CLIPSeg-based mask generator in Section III-A, and then presents the multimodal prompt design from three aspects, including morphological constraints in Section III-B, physical prior knowledge in Section III-C, and few-shot exemplar guidance in Section III-D.

A. Ground-Roll Mask Generator

Given a seismic gather (or its image-like representation) $I \in \mathbb{R}^{T\times X}$, we adopt CLIPSeg as the ground-roll mask generator due to its promptable dense prediction capability and its strong transferability from large-scale vision–language pretraining. CLIPSeg consists of a frozen vision–language backbone that aligns visual features with prompt embeddings, followed by a lightweight segmentation head that produces a prompt-conditioned dense response. Let $E_I(\cdot)$ denote the image encoder, $E_P(\cdot)$ the prompt encoder, and $D(\cdot,\cdot)$ the mask decoder [24]. The prompt-conditioned mask response can be expressed as

$$\widetilde{\mathbf{M}} = D(E_I(I),\, E_P(P))\,, \tag{5}$$

where P denotes the multimodal prompt set introduced in Sections III-B–III-D. The response map $\widetilde{\mathbf{M}} \in [0,1]^{T\times X}$ assigns each time–space sample a confidence value indicating its association with ground-roll patterns, with larger values corresponding to higher ground-roll dominance [32].

To obtain the binary support used in Stage II, the response map is converted as

$$M_{ij} = \begin{cases} 1, & \widetilde{M}_{ij} > \eta, \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

where η denotes a prescribed threshold. The resulting binary mask $\mathbf{M}$ indicates whether each time–space sample belongs to the ground-roll-dominant region, with $\mathbf{M}_{ij} = 1$ denoting ground-roll support and $\mathbf{M}_{ij} = 0$ otherwise.

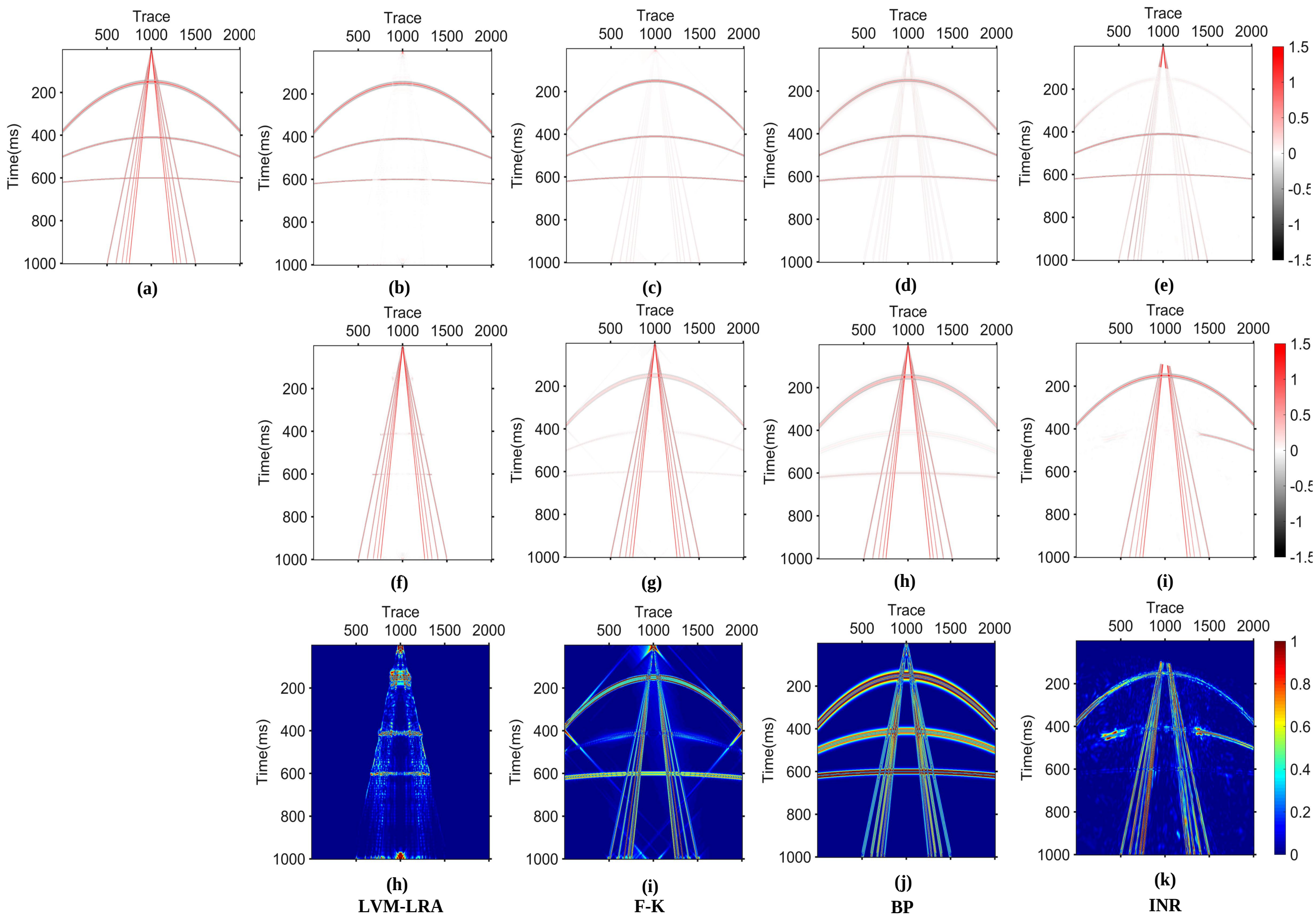


Fig. 2. Denoising comparison on synthetic VSP data. (a) Original VSP data . Denoised results, removal noise, and local similarity comparisons using (b)(f)(j) proposed LVM-LRA, (c)(g)(k) F-K, (d)(h)(l) BP, and (e)(i)(m) INR.

B. Prompt I: Morphological Constraints

ground-roll on a time–space gather typically exhibits distinctive visual patterns that can be recognized directly from its appearance. It often forms fan-shaped or cone-shaped energy distributions in near-offset regions and is usually expressed as laterally coherent, low-slope, and smoothly varying wavefronts [33, 34]. Motivated by these phenomenological observations, we encode such visible characteristics as morphological prompts to guide the vision–language model toward regions that *look like* ground-roll. Here, “morphology” refers to appearance-level cues rather than physical causes. That is, we emphasize how ground-roll is visually manifested in the gather, including its geometric shape, spatial continuity, and textural coherence, without explicitly invoking wave-propagation analysis or transform-domain representations such as f–k or τ–p. This design allows the prompt to remain lightweight, intuitive, and directly compatible with dense visual segmentation. Formally, we denote the morphological prompt set as

$$\mathcal{P}_m = \left\{ p_1^{(m)}, \ldots, p_{N_m}^{(m)} \right\}, \tag{7}$$

where each prompt captures one salient appearance attribute of ground-roll, such as *fan-/cone-shaped energy bands*, *low-slope coherent events*, or *smooth and continuous wavefront textures*. These prompts provide data-agnostic visual guidance that reduces semantic ambiguity under seismic domain shift and helps CLIPSeg generate a more spatially coherent dominance map M without task-specific finetuning [35]. Overall, morphological prompts primarily improve mask coverage and appearance consistency, thereby providing a stable spatial prior for Stage II.

C. Prompt II: Physical Prior Knowledge

Although morphological prompts describe what ground-roll *looks like*, appearance alone may not be sufficiently discriminative in complex seismic scenarios, where other coherent events can exhibit locally similar textures or geometric trends [36, 37]. To improve specificity and interpretability, we further introduce physics-informed prompts that describe why ground-roll exhibits such patterns from the perspective of wave generation and propagation. [38] pointed out that ground-roll can generally be regarded as a Rayleigh-type surface wave propagating along or near the ground surface, and is typically characterized by low velocity, low frequency, high amplitude, as well as guided-wave and dispersive behavior. These properties provide a causal layer of guidance beyond

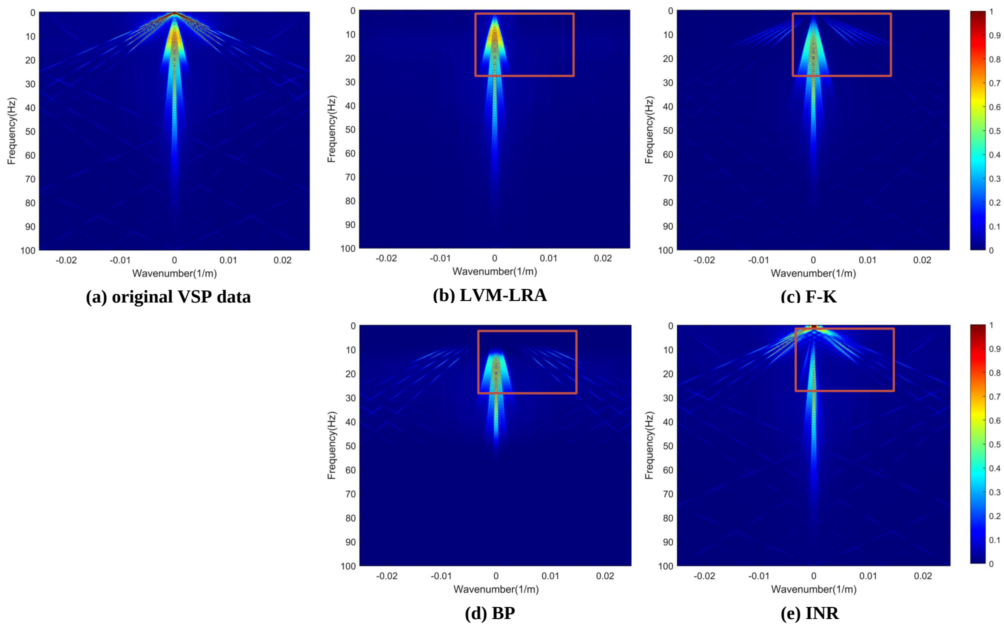


Fig. 3. Spectral comparison of the denoised results in Fig. 2: (a), (b), (c), (d), and (e), Frequency–wavenumber spectra of Fig. 2 (a), (b), (c), (d), (e), respectively.

incidental visual resemblance: ground-roll is not only a certain visual pattern, but also a specific type of near-surface, low-velocity, dispersive wave phenomenon. Accordingly, we define a physics-prior prompt set as

$$\mathcal{P}_p = \left\{ p_1^{(p)}, \ldots, p_{N_p}^{(p)} \right\}, \tag{8}$$

where each prompt encodes one interpretable physical attribute of ground-roll, such as *Rayleigh-type surface-wave propagation*, *low-velocity dispersive behavior*, or *near-surface guided-wave energy*. Different from $\mathcal{P}_m$, which emphasizes visible appearance, $\mathcal{P}_p$ highlights the underlying propagation mechanism and wave-property attributes of the target event. In practice, we combine $\mathcal{P}_m$ and $\mathcal{P}_p$ to form the multimodal prompt set P in (5), thereby obtaining a more reliable and interpretable ground-roll mask M under limited supervision [39].

D. Prompt III: Few-Shot Exemplar Guidance

While textual prompts provide high-level semantic guidance, some structural properties of seismic patterns remain difficult to express using natural language alone. To complement the textual description, we incorporate a single exemplar image into the multimodal prompt set P as a few-shot visual prompt. This exemplar serves as a lightweight cue to reduce ambiguity under seismic domain shift [40].

Specifically, a representative seismic image containing ground-roll events is selected as the visual exemplar. It implicitly conveys characteristic spatial patterns of ground-roll, such as coherent structures and fan-shaped geometry. During inference, the exemplar provides an additional visual anchor that helps CLIPSeg better associate textual descriptions with the actual appearance of ground-roll in seismic data [41, 42]. An important advantage of this strategy is that it requires only a single example image and does not involve additional training, making the overall pipeline lightweight and easy to deploy. By combining semantic text prompts with a few-shot visual exemplar, the proposed CLIPSeg-based masking framework becomes better adapted to seismic imagery while preserving the generalization capability of the pretrained vision–language representations.

TABLE I
ABLATION STUDY OF DIFFERENT PROMPT COMBINATIONS FOR GROUND-ROLL MASK GENERATION IN STAGE I.

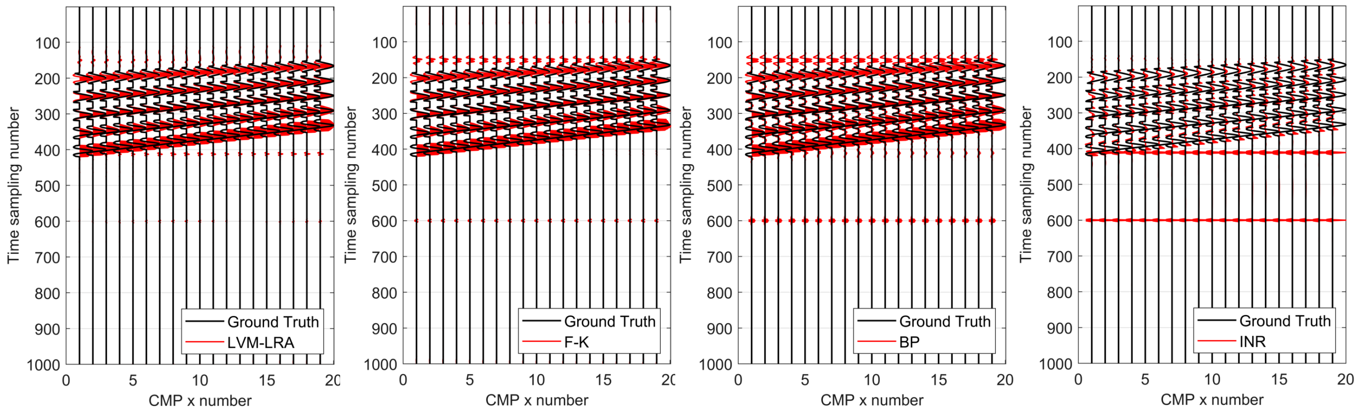
Prompt design	**Mask IoU**
Generic category-level text prompt ("ground-roll")	0.58
Morphological text prompt only	0.71
Physical-property text prompt only	0.76
Few-shot visual prompt only	0.87
Morphological + Physical text prompts	0.82
Morphological text + Visual prompt	0.89
Physical-property text + Visual prompt	0.90
Morphological + Physical + Visual prompts (proposed)	**0.92**

IV. STAGE II: MASKING-GUIDED LOW-RANK APPROXIMATION FOR GROUND-ROLL SUPPRESSION

Stage II performs ground-roll suppression under the guidance of the mask generated in Stage I. This section first introduces the mask-guided low-rank formulation for reflection and ground-roll separation in Section IV-A, and then presents the corresponding ADMM-based optimization procedure in Section IV-B.

A. Mask-Guided Low-Rank Formulation

In accordance with the general formulation in (4), we adopt nuclear norms for the reflection and ground-roll components,



(a) LVM-LRA **(b) F-K** **(c) BP** **(d) INR**

Fig. 4. Single-trace comparison of the ground-roll at the 800-818th trace between the original ground-roll and the extracted components by (a) proposed LVM-LRA, (b) F-K, (c) BP, and (d) INR.

Algorithm 1 LVM-LRA for Ground-Roll Attenuation

Input: Observed gather $\mathbf{Y} \in \mathbb{R}^{T\times X}$; threshold η; ADMM parameters $(\lambda_{\rm g}, \lambda_{\rm s}, \rho, K_m, \varepsilon_{\rm p}, \varepsilon_{\rm f})$.
Output: Reflection component $\widehat{\mathbf{X}}$, ground-roll component $\widehat{\mathbf{G}}$

Stage I: LVM-guided mask generation
Construct image-like representation I from $\mathbf{Y}$
Construct multimodal prompt set P
$\widetilde{\mathbf{M}} \leftarrow D(E_I(I),\, E_P(P))$
Binarize $\widetilde{\mathbf{M}}$ using threshold η to obtain $\mathbf{M}$
Stage II: Mask-guided low-rank decomposition
Initialize $\mathbf{X}^0, \mathbf{G}^0, \mathbf{J}^0, \mathbf{K}^0, \mathbf{\Lambda}_G^0, \mathbf{\Lambda}_X^0 \leftarrow \mathbf{0}$, $k \leftarrow 0$
while not converged and $k < K_{\max}$ **do**
  Update $\mathbf{X}^{k+1}$ using (13)
  Update $\widetilde{\mathbf{G}}^{k+1}$ using (15)
  Update $\mathbf{G}^{k+1}$ using (16)
  Update $\mathbf{J}^{k+1}$ using (18)
  Update $\mathbf{K}^{k+1}$ using (20)
  Update $\mathbf{\Lambda}_G^{k+1}$ and $\mathbf{\Lambda}_X^{k+1}$ using (23)–(24)
  Check convergence using (25)
  $k \leftarrow k+1$
end while
$\widehat{\mathbf{X}} \leftarrow \mathbf{X}^{k+1}, \quad \widehat{\mathbf{G}} \leftarrow \mathbf{G}^{k+1}$

leading to

$$\min_{\mathbf{X},\mathbf{G}} \frac{1}{2}\|\mathbf{Y}-\mathbf{X}-\mathbf{G}\|_F^2 + \lambda_{\rm s}\|\mathbf{X}\|_* + \lambda_{\rm g}\|\mathbf{G}\|_*, \tag{9}$$

subject to the mask support constraint in (3), where $\lambda_{\rm g}$ and $\lambda_{\rm s}$ are the regularization parameters for the ground-roll and reflection components, respectively.

In this formulation, the data-fidelity term enforces that the estimated components jointly explain the observed seismic gather, while the two nuclear-norm terms promote structured and predictable representations for both $\mathbf{G}$ and $\mathbf{X}$ [43]. For the ground-roll component, the low-rank prior is motivated by its strong spatial correlation and local predictability in the time–space domain. For the reflection component, low-rank regularization helps preserve event continuity and suppress residual fluctuations that are not consistent with coherent reflection structures [25, 44, 45].

Compared with a purely global low-rank decomposition, the present model incorporates the mask support constraint from Stage I, so that the ground-roll component is estimated only within the detected ground-roll-dominant region [31]. As a result, the low-rank prior imposed on $\mathbf{G}$ becomes spatially selective rather than globally applied, which improves the controllability of the separation and reduces the risk of assigning reflection energy outside the mask to the ground-roll component.

B. Optimization via ADMM

To solve (9), we employ the alternating direction method of multipliers (ADMM), in which one variable is optimized at a time while the others are kept fixed. By introducing two auxiliary variables $\mathbf{J}$ and $\mathbf{K}$, the problem in (9) can be reformulated as

$$\begin{aligned} \min_{\mathbf{X},\mathbf{G},\mathbf{J},\mathbf{K}} \quad & \frac{1}{2}\|\mathbf{Y}-\mathbf{X}-\mathbf{G}\|_F^2 + \lambda_{\rm g}\|\mathbf{J}\|_* + \lambda_{\rm s}\|\mathbf{K}\|_*, \\ \text{s.t.} \quad & \mathbf{G}=\mathbf{J}, \qquad \mathbf{X}=\mathbf{K}, \end{aligned} \tag{10}$$

where the mask support constraint in (3) is enforced in the update of $\mathbf{G}$. Using the scaled-form ADMM, the augmented Lagrangian is written as

$$\begin{aligned} &\mathcal{L}_\rho(\mathbf{X},\mathbf{G},\mathbf{J},\mathbf{K},\mathbf{\Lambda}_G,\mathbf{\Lambda}_X) \\ &= \frac{1}{2}\|\mathbf{Y}-\mathbf{X}-\mathbf{G}\|_F^2 + \lambda_{\rm g}\|\mathbf{J}\|_* + \lambda_{\rm s}\|\mathbf{K}\|_* \\ &\quad + \frac{\rho}{2}\|\mathbf{G}-\mathbf{J}+\mathbf{\Lambda}_G\|_F^2 - \frac{\rho}{2}\|\mathbf{\Lambda}_G\|_F^2 \\ &\quad + \frac{\rho}{2}\|\mathbf{X}-\mathbf{K}+\mathbf{\Lambda}_X\|_F^2 - \frac{\rho}{2}\|\mathbf{\Lambda}_X\|_F^2, \end{aligned} \tag{11}$$

where $\mathbf{\Lambda}_G$ and $\mathbf{\Lambda}_X$ are the scaled dual variables and $\rho > 0$ is the penalty parameter.

1) Updating $\mathbf{X}_{k+1}$: With the other variables fixed, the $\mathbf{X}$-related subproblem is

$$\mathbf{X}_{k+1} = \arg\min_{\mathbf{X}} \frac{1}{2}\|\mathbf{Y}-\mathbf{X}-\mathbf{G}_k\|_F^2 + \frac{\rho}{2}\|\mathbf{X}-\mathbf{K}_k+\mathbf{\Lambda}_{X,k}\|_F^2. \tag{12}$$

Since (12) is a quadratic problem with respect to $\mathbf{X}$, its closed-form solution is given by

$$\mathbf{X}_{k+1} = \frac{(\mathbf{Y}-\mathbf{G}_k) + \rho(\mathbf{K}_k - \mathbf{\Lambda}_{X,k})}{1+\rho}. \tag{13}$$

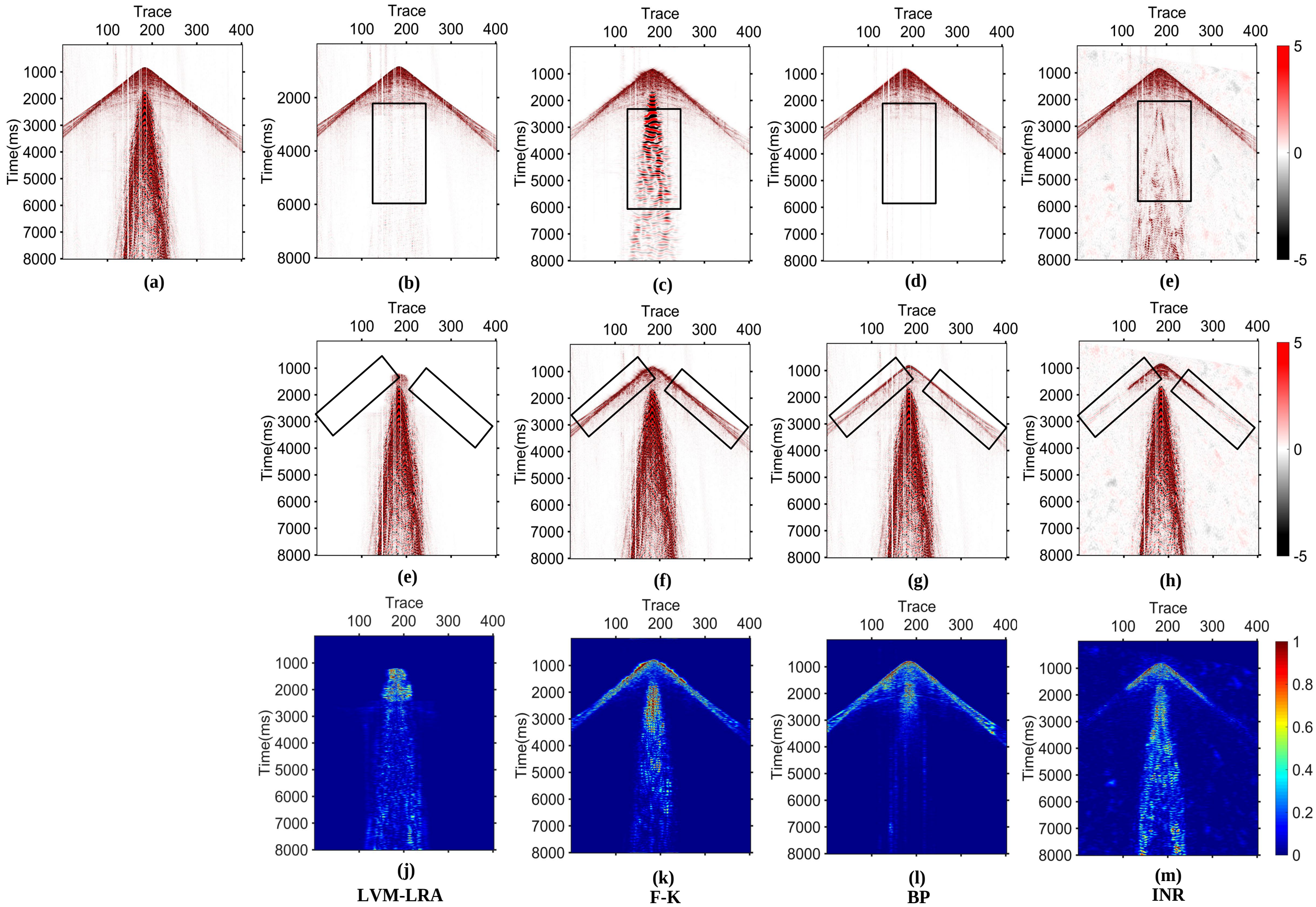


Fig. 5. Denoising comparison on the second real VSP data. (a) Original VSP data. Denoised results, removal noise, and local similarity comparisons using (b)(f)(j) proposed LVM-LRA, (e)(g)(k) F-K, (d)(h)(l) BP, and (e)(i)(m) INR.

This step updates the reflection-dominant component from the current residual while keeping it close to its low-rank proxy $\mathbf{K}$.

2) Updating $\mathbf{G}_{k+1}$*:* Similarly, the $\mathbf{G}$-related subproblem is

$$\mathbf{G}_{k+1} = \arg\min_{\mathbf{G}} \frac{1}{2}\|\mathbf{Y}-\mathbf{X}_{k+1}-\mathbf{G}\|_F^2+\frac{\rho}{2}\|\mathbf{G}-\mathbf{J}_k+\mathbf{\Lambda}_{G,k}\|_F^2, \tag{14}$$

subject to the mask support constraint in (3). Ignoring the mask constraint temporarily, the unconstrained minimizer can be written as

$$\widetilde{\mathbf{G}}_{k+1} = \frac{(\mathbf{Y}-\mathbf{X}_{k+1})+\rho(\mathbf{J}_k-\mathbf{\Lambda}_{G,k})}{1+\rho}. \tag{15}$$

Then, the mask support prior is enforced through the projection

$$\mathbf{G}_{k+1} = \mathbf{M} \circ \widetilde{\mathbf{G}}_{k+1}. \tag{16}$$

Hence, the estimated ground-roll energy is restricted to the detected ground-roll-dominant region, while the energy outside the mask is naturally retained in the reflection component.

3) Updating $\mathbf{J}_{k+1}$*:* The $\mathbf{J}$-related subproblem is given by

$$\mathbf{J}_{k+1} = \arg\min_{\mathbf{J}} \lambda_{\mathrm{g}}\|\mathbf{J}\|_* + \frac{\rho}{2}\|\mathbf{G}_{k+1}-\mathbf{J}+\mathbf{\Lambda}_{G,k}\|_F^2. \tag{17}$$

This subproblem corresponds to the proximal mapping of the nuclear norm and can be solved by singular value thresholding (SVT), i.e.,

$$\mathbf{J}_{k+1} = \mathrm{SVT}_{\lambda_{\mathrm{g}}/\rho}(\mathbf{G}_{k+1}+\mathbf{\Lambda}_{G,k}). \tag{18}$$

4) Updating $\mathbf{K}_{k+1}$*:* Likewise, the $\mathbf{K}$-related subproblem is

$$\mathbf{K}_{k+1} = \arg\min_{\mathbf{K}} \lambda_{\mathrm{s}}\|\mathbf{K}\|_* + \frac{\rho}{2}\|\mathbf{X}_{k+1}-\mathbf{K}+\mathbf{\Lambda}_{X,k}\|_F^2. \tag{19}$$

Its analytical solution is also given by the SVT operator:

$$\mathbf{K}_{k+1} = \mathrm{SVT}_{\lambda_{\mathrm{s}}/\rho}(\mathbf{X}_{k+1}+\mathbf{\Lambda}_{X,k}). \tag{20}$$

For completeness, let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the singular value decomposition of $\mathbf{A}$. Then the SVT operator is defined as

$$\mathrm{SVT}_\tau(\mathbf{A}) = \mathbf{U}\,\mathcal{S}_\tau(\mathbf{\Sigma})\,\mathbf{V}^\top, \tag{21}$$

where $\mathcal{S}_\tau(\mathbf{\Sigma})$ applies soft-thresholding to the singular values:

$$\mathcal{S}_\tau(\sigma_i) = \max(\sigma_i - \tau,\, 0). \tag{22}$$

5) Updating $\mathbf{\Lambda}_{G,k+1}$ *and* $\mathbf{\Lambda}_{X,k+1}$*:* After updating the primal variables, the scaled dual variables are updated as

$$\mathbf{\Lambda}_{G,k+1} = \mathbf{\Lambda}_{G,k} + \mathbf{G}_{k+1} - \mathbf{J}_{k+1}, \tag{23}$$

$$\mathbf{\Lambda}_{X,k+1} = \mathbf{\Lambda}_{X,k} + \mathbf{X}_{k+1} - \mathbf{K}_{k+1}. \tag{24}$$

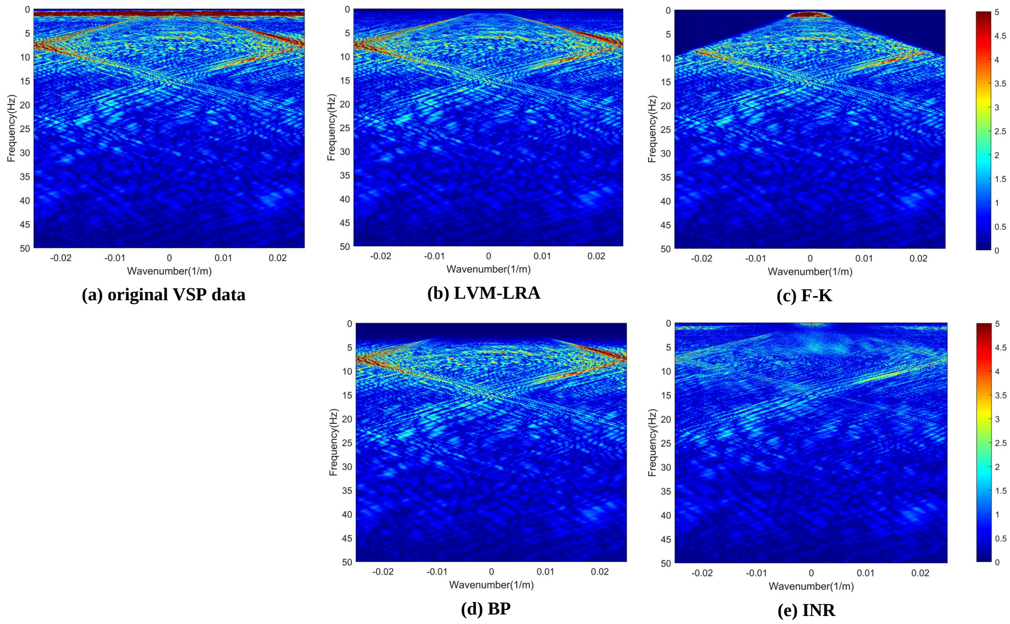


Fig. 6. Spectral comparison of the denoised results in Fig. 5: (a), (b), (c), (d), and (e) Frequency–wavenumber spectra of Fig. 5 (a), (b), (c), (d), and (e), respectively.

The iterations are terminated when the primal consistency residuals and the data-fitting residual become sufficiently small, namely,

$$\begin{aligned} \|\mathbf{G}_k - \mathbf{J}_k\|_F &\le \varepsilon_{\mathrm{p}}, \\ \|\mathbf{X}_k - \mathbf{K}_k\|_F &\le \varepsilon_{\mathrm{p}}, \\ \|\mathbf{Y} - \mathbf{X}_k - \mathbf{G}_k\|_F &\le \varepsilon_{\mathrm{f}}, \end{aligned} \tag{25}$$

or when a preset maximum number of iterations is reached. In practice, λ_{g} and λ_{s} control the balance between ground-roll suppression and reflection preservation, whereas ρ mainly affects the convergence behavior.

C. Computational Complexity

We analyze the computational cost of the proposed Stage II solver in terms of the gather size $T \times X$, the target truncated rank $r \ll \min(T, X)$ used in the SVT updates, and the number of ADMM iterations K.

Per-iteration cost: The updates of $\mathbf{X}$ and $\mathbf{G}$ in (13) and (15) are dominated by element-wise matrix additions and scalings, requiring $\mathcal{O}(TX)$ operations. Enforcing the binary mask support via the projection $\mathbf{G} \leftarrow \mathbf{M} \circ \widetilde{\mathbf{G}}$ in (16) also costs $\mathcal{O}(TX)$. The dual updates in (23)–(24) and residual evaluations in (25) are likewise $\mathcal{O}(TX)$. The dominant cost arises from the two SVT steps in (18) and (20), each requiring a singular value decomposition (SVD) of a $T \times X$ matrix. In practice, we employ truncated SVD with rank r for efficiency. Using Krylov-subspace or randomized SVD implementations, each truncated SVD typically costs on the order of $\mathcal{O}(TXr)$ (with additional lower-order terms such as $\mathcal{O}((T + X)r^2)$ depending on the specific routine). Therefore, the overall per-iteration complexity is dominated by

$$\mathcal{O}\big(TXr\big) \quad + \mathcal{O}(TX) \quad . \tag{26}$$

Total complexity: With K ADMM iterations, the total time complexity can be summarized as $\mathcal{O}\big(KTXr\big)$, where the $\mathcal{O}(KTX)$ terms from the remaining updates are lower-order when $r \ll \min(T, X)$. The memory footprint is $\mathcal{O}(TX)$ to store the primal/dual variables ($\mathbf{X}, \mathbf{G}, \mathbf{J}, \mathbf{K}$ and $\mathbf{\Lambda}_X, \mathbf{\Lambda}_G$) and the mask $\mathbf{M}$, while the truncated SVD requires additional workspace of roughly $\mathcal{O}((T + X)r)$ for the leading singular vectors.

V. Experimental Setup

A. Baselines

Following the literature review in Section I, three representative seismic noise attenuation methods are selected as baselines. Their core denoising principles are briefly summarized as follows.

1) *Frequency–Wavenumber (F–K) Filtering [46]:* F–K filtering exploits differences in apparent velocity among wavefield components. The seismic data are transformed into the frequency–wavenumber domain, where ground-roll energy concentrated in low-frequency and low-velocity regions is attenuated prior to inverse transformation.
2) *Bandpass Filtering [47]:* Bandpass filtering suppresses noise by restricting the effective frequency range of the data. By exploiting the predominantly low-frequency characteristics of ground roll, frequency components outside the target signal band are attenuated to reduce low-frequency noise energy.
3) *Implicit Neural Representation (INR) [4]:* INR represents seismic data as a continuous implicit function that maps spatial–temporal coordinates to amplitudes. By fitting this function to the observed data, dominant structured signal components can be captured, whereas noise components that are inconsistent with the implicit

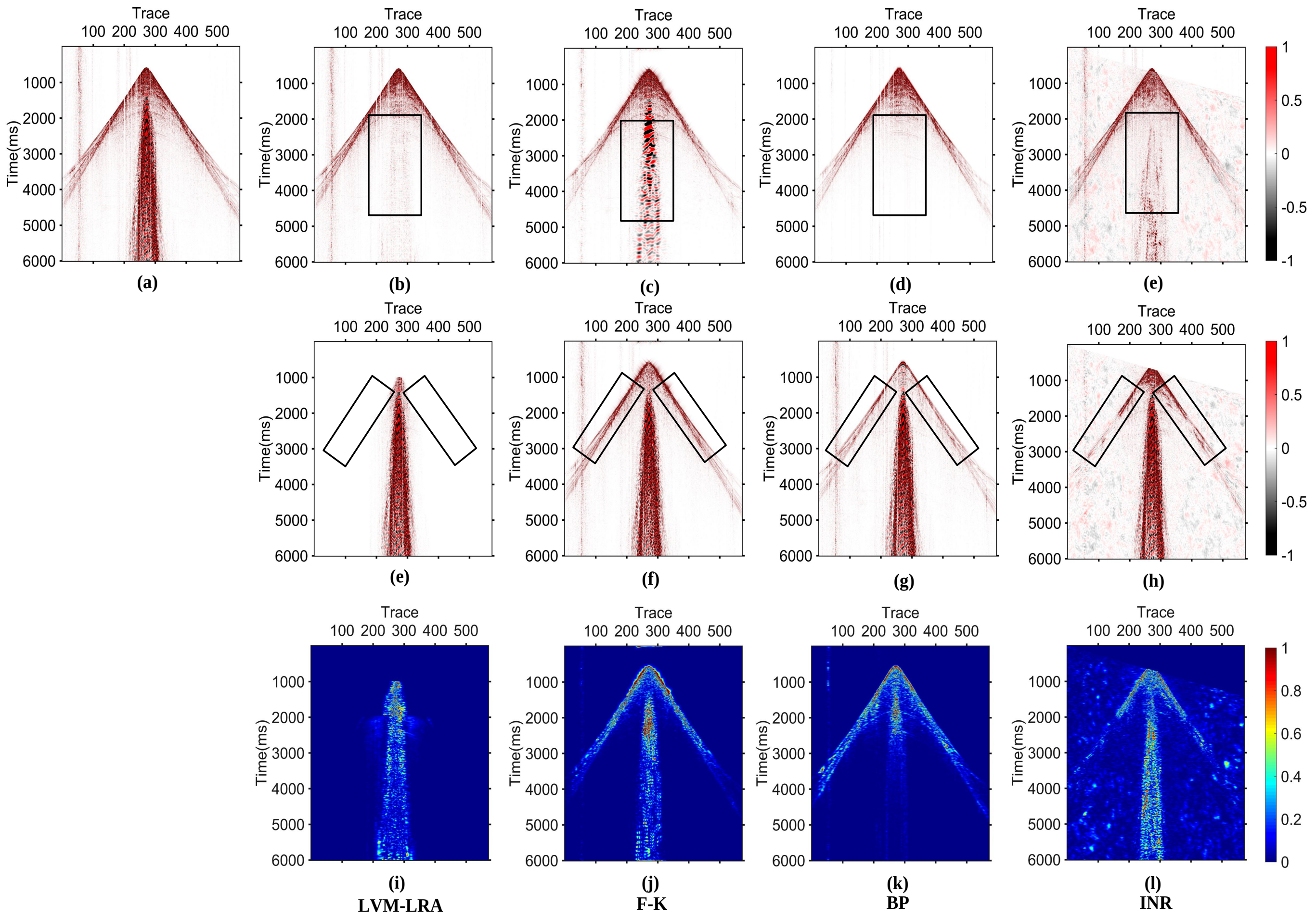


Fig. 7. Denoising comparison on the first real VSP data. (a) Original VSP data. Denoised results, removal noise, and local similarity comparisons using (b)(f)(j) proposed LVM-LRA, (e)(g)(k) F-K, (d)(h)(l) BP, and (e)(i)(m) INR.

representation tend to be suppressed during optimization.

To ensure a fair comparison and reliable performance evaluation, the baseline methods are implemented by following the experimental settings and parameter configurations reported in [46], [47], and [4]. Only minor adjustments are made when necessary to accommodate the characteristics of the evaluated datasets.

B. Evaluation Metrics

Following the methodologies outlined in [48] and [49], signal-to-noise ratio (SNR) and local similarity are adopted as quantitative metrics to assess the denoising performance of the proposed LVM-LRA method against existing SOTA approaches. For synthetic datasets, the availability of ground-truth data enables objective evaluation using SNR, which quantifies the proportion of the preserved VSP signal relative to residual ground-roll and random noise.

For real VSP datasets, where ground truth is unavailable, the performance of the proposed LVM-LRA method is assessed using local similarity instead of SNR. The methodology and implementation of this metric are described in [50] and [51]. Importantly, local similarity is computed at appropriate spatial scales to assess signal leakage. This is typically achieved by comparing the denoised result with the extracted ground-roll component: when the extracted noise contains residual signal energy, elevated similarity values appear in the corresponding regions, indicating incomplete separation.

C. Hyperparameter Configurations

After establishing the LVM-LRA framework described in Section II, two categories of parameters must be specified prior to implementation: the Stage I parameters associated with input representation and prompt aggregation for LVM-based segmentation, and the Stage II parameters associated with the mask-guided low-rank decomposition, including the nuclear-norm weights and the ADMM solver settings.

Specifically, in Stage I, a hybrid prompting strategy is adopted to improve the robustness of mask estimation. This strategy consists of two text prompts and one few-shot visual prompt, where $N_t = 2$ corresponds to the morphological prompt and the physical-property prompt, and $N_e = 1$ denotes a representative visual exemplar of ground-roll. In this study, the two text prompts are specified as “fan-shaped, low-frequency, coherent seismic events near the surface” and

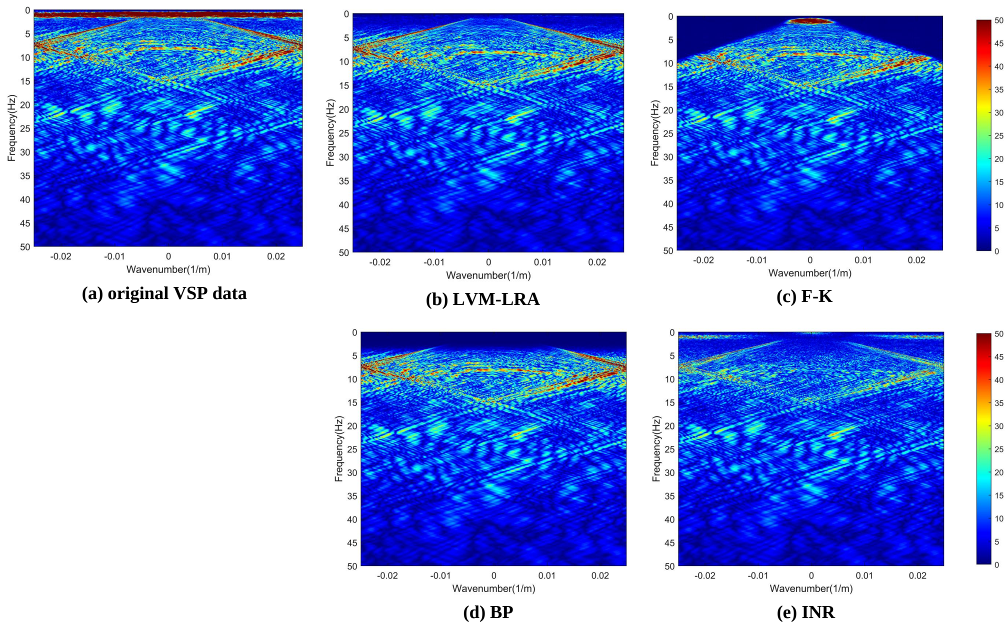


Fig. 8. Spectral comparison of the denoised results in Fig. 7: (a), (b), (c), (d), and (e) Frequency–wavenumber spectra of Fig. 7 (a), (b), (c), (d), and (e), respectively.

"surface-wave noise with low apparent velocity and dispersion," respectively. The predictions produced by all prompts are then aggregated to generate the mask M.

In Stage II, the masking constraint is enforced as $G = M \circ G$. As shown in (9), the nuclear-norm regularization weights are set to $\lambda_{\mathrm{g}} = 1.0 \times 10^{-2}$ and $\lambda_{\mathrm{s}} = 5.0 \times 10^{-3}$ after normalizing Y. The ADMM penalty parameter is set to $\rho = 1$, and the maximum number of iterations is set to $K_{\max} = 200$. The stopping tolerances are specified as $\varepsilon_{\mathrm{p}} = 10^{-4}$ and $\varepsilon_{\mathrm{f}} = 10^{-5}$. Unless otherwise specified, the same parameter settings are used for all experiments.

D. Ablation Studies

Ablation studies are conducted on synthetic datasets to evaluate the contribution of the proposed few-shot LVM-guided masking strategy. As discussed in the problem formulation, ground-roll masks can be obtained either through rule-based hard-thresholding schemes [29] or through supervised segmentation networks trained with pixel-level annotations [30, 31]. In this work, supervised deep-learning-based mask predictors are not included as ablation baselines, because they require additional training procedures and extensive labeled data, which are inconsistent with the training-free setting of the proposed framework.

Instead, the proposed method adopts a few-shot prompting strategy, in which two text prompts and one representative ground-roll image exemplar are used to guide the LVM in producing a response map without task-specific training. This response map is subsequently converted into a binary mask through hard thresholding. Accordingly, the ablation study is designed to compare the proposed few-shot LVM-guided binary mask with conventional rule-based hard-threshold masks, thereby isolating the effect of semantic prompt guidance on mask generation.

VI. Results

A. Validation on Synthetic VSP Data

For the synthetic experiment, a clean VSP reference is used, and a strong ground-roll component is superimposed to generate the contaminated input with an SNR of 1.45 dB, as shown in Fig. 2(a). Figure 2 summarizes the results obtained by different methods, including the reconstructed VSP signal sections, the extracted ground-roll components, and the local similarity maps between the reconstructed signals and the extracted ground-roll. The proposed LVM-LRA method is compared with three representative baselines, namely F–K, BP, and INR. From the recovered VSP sections (Figs. 2(b)–(e)) together with the corresponding extracted ground-roll sections (Figs. 2(f)–(i)), LVM-LRA exhibits the most coherent reflection continuity while effectively removing the dominant ground-roll fan.

Additional evidence is provided by the local similarity maps in the bottom row of Fig. 2. Since the clean reference contains only the desired reflection events, high similarity aligned with reflection moveout indicates better signal preservation, whereas abnormal similarity patterns concentrated within the ground-roll-dominant cone suggest residual interference or incomplete separation. In this regard, the similarity map produced by LVM-LRA shows responses concentrated along the main reflection events, with reduced similarity inside the ground-roll cone, which is consistent with the intended separation behavior. By contrast, the baseline methods produce broader or more fragmented similarity structures in the central overlap zone, reflecting less stable separation under strong overlap between ground-roll and reflections.

The advantage of LVM-LRA is further corroborated in the frequency–wavenumber domain, as shown in Fig. 3. The spectrum of the original data (Fig. 3(a)) exhibits pronounced low-velocity ground-roll energy distributed away from the

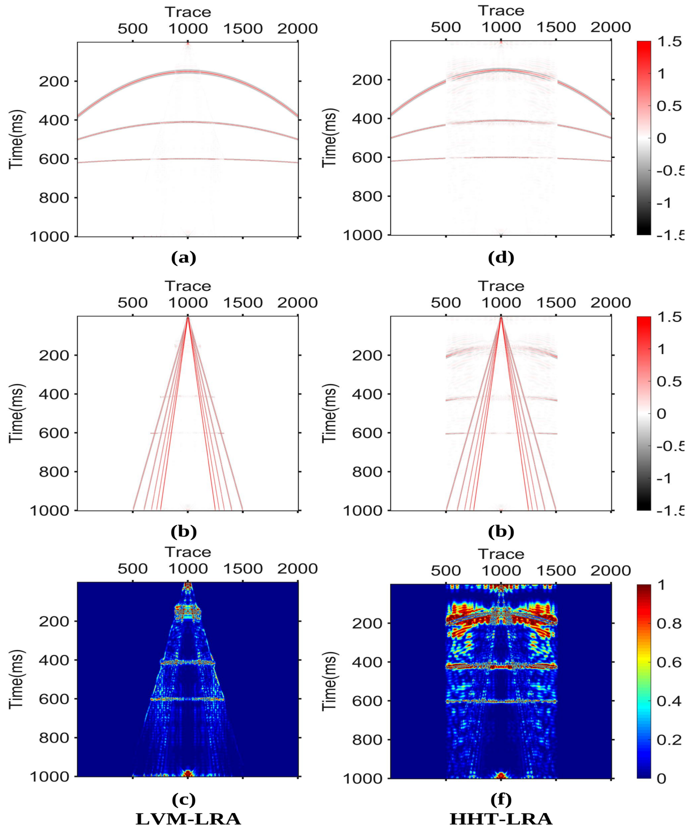


Fig. 9. Results of the ablation studies. Denoised results, removal noise, and local similarity comparisons using (a)(c)(e) proposed LVM-LRA, (b)(d)(f) HHT-LRA.

compact reflection band. After processing, the LVM-LRA spectrum (Fig. 3(b)) becomes more concentrated, with noticeably reduced ground-roll-related dispersion, whereas the spectra of F–K, BP, and INR (Figs. 3(c)–(e)) retain more dispersed components associated with residual ground-roll energy. This transform-domain analysis provides an interpretable complement to the section-based comparisons shown in Fig. 2.

To complement the qualitative analysis, quantitative evaluations are reported in the accompanying tables using the metrics defined in Section V-B. Overall, LVM-LRA achieves the most favorable balance between ground-roll attenuation and reflection preservation, which is consistent with both the visual comparisons in Fig. 2 and the spectral evidence in Fig. 3. In addition, a trace-level comparison is presented in Fig. 4 to assess waveform fidelity within the reflection-dominant window. The traces reconstructed by LVM-LRA (red) closely follow the ground truth (black) in both event timing and moveout-consistent waveform shape. In contrast, the baseline methods (F–K, BP, and INR) exhibit more pronounced spurious bumps and spikes on several traces, indicating residual ground-roll interference and/or waveform distortion.

B. Validation on the First Real VSP Data

To further evaluate the robustness of the proposed LVM-LRA framework, a series of experiments are conducted on the first real VSP dataset. Since no clean reference is available for field data, the evaluation relies on visual inspection of the reconstructed signal and extracted ground-roll sections, together with local similarity maps that highlight potential signal leakage in ground-roll-dominant regions.

Figure 5 presents the processing results of LVM-LRA and three baselines, namely F–K, BP, and INR. Specifically, Figs. 5(b)–(e) show the reconstructed VSP sections, Figs. 5(f)–(i) show the corresponding extracted ground-roll components, and Figs. 5(j)–(m) present the local similarity maps computed between the reconstructed signal and the extracted noise. In the absence of ground truth, structured similarity responses aligned with reflection events indicate signal leakage into the extracted noise, whereas reduced similarity within the ground-roll cone suggests cleaner separation. As shown in Figs. 5(b), (e), and (j), LVM-LRA yields a reconstructed section with clearer reflection continuity, while the extracted noise is concentrated mainly within the ground-roll fan. Moreover, its local similarity map exhibits weaker and less structured responses in the central overlap zone, indicating improved separation between reflections and ground-roll.

Further evidence is provided in the frequency–wavenumber domain in Fig. 6, which compares the spectra of the original field data and the results produced by different methods. The f–k spectra provide an interpretable view of the ground-roll energy distribution in the field record. After processing, the LVM-LRA spectrum becomes more concentrated, with reduced dispersion associated with low-velocity ground-roll energy. This transform-domain behavior is consistent with the section-based observations in Fig. 5.

C. Validation on the Second Real VSP Data

To further examine the adaptability of the proposed LVM-LRA framework to different structural characteristics of field data, additional experiments are conducted on the second real VSP dataset, in which the reflection events exhibit a relatively narrower dip range. Figure 7 presents the denoising comparison on this second field record. The original VSP section is shown in Fig. 7(a), while the reconstructed signal sections obtained by LVM-LRA, F–K, BP, and INR are displayed in Figs. 7(b)–(e). The corresponding removed components, interpreted as extracted ground-roll/noise, are presented in Figs. 7(f)–(i). To diagnose potential signal leakage in the absence of ground truth, local similarity is computed between the reconstructed signal and the removed component, and the resulting maps are shown in Figs. 7(j)–(m). Under this definition, structured similarity responses aligned with reflection events indicate that coherent signal energy has leaked into the removed component, whereas weaker and less structured similarity within the ground-roll-dominant region suggests cleaner separation.

Further evidence is provided in the frequency–wavenumber domain in Fig. 8. The spectrum of the original field data (Fig. 8(a)) reveals the characteristic distribution of low-velocity ground-roll energy. After processing, the LVM-LRA spectrum (Fig. 8(b)) becomes more concentrated, with reduced dispersion associated with ground-roll interference. This transform-domain behavior is consistent with the time–space-

TABLE II
PERFORMANCE COMPARISON OF SYNTHETIC VSP DATA BASED ON SNR AND LOCAL SIMILARITY METRICS

Methods	SNR(dB)	Local Similarity	
		Average	Variance
INR	-03.34	0.0834	0.0352
BP	04.21	0.1268	0.0688
F-K	07.54	0.0897	0.0397
HHT-LRA	07.76	0.0971	0.0445
Proposed LVM-LRA	**14.50**	**0.0285**	**0.0100**

TABLE III
PERFORMANCE COMPARISON OF REAL VSP DATA BASED ON LOCAL SIMILARITY METRICS

Methods	Real VSP Data I		Real VSP Data II	
	Average	Variance	Average	Variance
INR	0.0493	0.0164	0.2179	0.0635
BP	0.0360	0.0143	0.1026	0.0335
F-K	0.0481	0.0111	0.1571	0.0415
Proposed LVM-LRA	**0.0274**	**0.0094**	**0.0333**	**0.0136**

domain observations in Fig. 7. The spectra produced by F–K, BP, and INR are shown in Figs. 8(c)–(e) for reference.

D. Discussion

1) Ablation Studies: Ablation experiments are conducted on the synthetic dataset to isolate the effect of the masking strategy used to guide the low-rank decomposition. Figure 9 summarizes the qualitative comparison between the proposed LVM-LRA framework and a hard-threshold-based variant (HHT-LRA). Specifically, Figs. 9(a)–(b) show the reconstructed signal sections, Figs. 9(c)–(d) present the removed components interpreted as extracted ground-roll (noise), and Figs. 9(e)–(f) report the corresponding local similarity maps for diagnosing potential signal leakage. Since a clean reference is available in the synthetic case, SNR is further adopted for quantitative evaluation. The SNR scores reported in Table II provide an objective complement to the visual comparisons in Fig. 9, thereby enabling a controlled assessment of how different masking strategies affect signal preservation and ground-roll attenuation.

2) Limitation: Although the proposed LVM-LRA framework demonstrates strong performance on the evaluated datasets, it should be noted that the ground-roll energy in these data spans a relatively wide dip range, which allows partial separability from reflection events in the transform domain. In more challenging scenarios, where the ground-roll occupies a narrower dip range and exhibits stronger overlap with reflections, the separation problem becomes substantially more ill-posed [52].

This limitation is fundamentally associated with the reduced structural distinguishability between signal and noise, which affects not only transform-based methods [53] but also learning-based approaches [54]. Under such conditions, the effectiveness of the mask-guided low-rank decomposition may depend more critically on the accuracy of the estimated support region.

Nevertheless, the proposed LVM-guided masking strategy offers a promising direction for handling strongly overlapping interference, as it introduces semantic priors that are not restricted to purely transform-domain separability. Future work will focus on improving the robustness of mask estimation under severe overlap conditions and extending the framework to more challenging field scenarios.

VII. CONCLUSION

In this work, we introduce an LVM-LRA framework for ground-roll suppression in VSP gathers, formulated as a training-free separation approach. The key idea is to first obtain a promptable soft mask from a LVM, where the seismic gather is converted into a visual representation and the ground-roll-dominant region is identified via text prompts and a lightweight exemplar reference, without learning a dedicated segmentation network. This soft mask is then embedded into a mask-guided low-rank decomposition stage, where the signal and ground-roll components are estimated under physically motivated constraints and low-rank priors. The resulting optimization is solved efficiently using an ADMM-based procedure with fixed hyperparameter settings, enabling stable separation while avoiding the need for pixel-level annotations or paired clean/noisy training data. Extensive experiments on synthetic and real VSP datasets demonstrate that the proposed LVM-LRA method achieves a favorable balance between ground-roll attenuation and signal preservation, consistently outperforming representative baselines such as F–K filtering, bandpass filtering, and INR in both section-based evaluations and transform-domain analyses, and showing improved robustness in challenging overlap regions.

Future research will extend the proposed framework to broader geophysical scenarios, including multi-component VSP/DAS noise suppression, multi-event wavefield separation, and 3D/4D acquisition settings where spatial variability and nonstationarity are more severe. Beyond application-level extensions, we plan to explore more adaptive and scalable masking-and-decomposition designs, such as prompt optimization and uncertainty-aware mask aggregation, multi-scale or tensorized low-rank models, and accelerated solvers (e.g., warm-started ADMM and GPU-friendly proximal updates) to further improve robustness, efficiency, and generalization under more complex acquisition geometries and stronger ground-roll interference.

REFERENCES

[1] R. Saatcilar and N. Canitez, “A method of ground-roll elimination,” *Geophysics*, vol. 53, no. 7, pp. 894–902, Jul. 1988.

[2] R. Askari and H. R. Siahkoohi, “Ground roll attenuation using the s and x-f-k transforms,” *Geophysical Prospecting*, vol. 56, no. 1, pp. 105–114, Jan. 2008.

[3] L. Yang, S. Wang, X. Chen, O. M. Saad, W. Cheng, and Y. Chen, “Deep learning with fully convolutional and dense connection framework for ground roll attenuation,” *Surv. Geophys.*, vol. 44, no. 6, pp. 1919–1952, Mar 2023.
[4] J. Li, T. Alkhalifah *et al.*, “Unsupervised ground-roll attenuation via implicit neural representations,” *Geophysics*, vol. 90, no. 2, pp. V111–V121, Feb. 2025.
[5] D. C. Henley, “Coherent noise attenuation in the radial trace domain,” *Geophysics*, vol. 68, no. 4, pp. 1408–1416, Jan. 2003.
[6] F. Leite, R. Montagne, G. Corso, and L. Lucena, “Seismic entangled patterns analyzed via multiresolution decomposition,” *Nonlinear Processes Geophys.*, vol. 16, no. 2, pp. 211–217, May 2009.
[7] H. Li, W. Yang, and X. Yong, “Deep learning for ground-roll noise attenuation,” in *SEG International Exposition and Annual Meeting*. SEG, Oct. 2018, pp. SEG–2018.
[8] S. Sun, H. Li, G. Li, and Z. Wang, “Ground roll attenuation in seismic data based on enhanced deep learning framework with adaptive frequency modulation loss,” *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–16, Jul. 2024.
[9] Y. Hu, L. Wang, F. Cheng, Y. Luo, C. Shen, and B. Mi, “Ground-roll noise extraction and suppression using high-resolution linear radon transform,” *J. Appl. Geophy.*, vol. 128, pp. 8–17, May 2016.
[10] S. Zhang, G. Wang, L. Lu, M. Wei, X. Qin, and L. Zhang, “Tf-unet: a time–frequency feature network for seismic noise attenuation using hybrid synthetic–field data: S. zhang et al.” *J. Supercomputing*, vol. 81, no. 18, p. 1630, Dec. 2025.
[11] J. Ma, K. Zhao, and Z. Liao, “A three-parameter high-resolution 3d fast radon transform,” *J. Geophy. Eng.*, vol. 22, no. 2, pp. 465–478, Sep. 2025.
[12] H. Lin, L. Xing, Q. Li, H. Liu, H. Zhang, and H. Zhou, “Spatial-domain synchrosqueezing wavelet transform and its application to seismic ground roll suppression,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–16, Sep. 2022.
[13] Z. Liu, Y. Chen, and J. Ma, “Ground roll attenuation by synchrosqueezed curvelet transform,” *J. Appl. Geophy.*, vol. 151, pp. 246–262, Apr. 2018. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0926985117309795
[14] Z. Feng, “Complex noise suppression using a robust dictionary learning approach,” *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, Dec. 2024.
[15] Y. Li, H. Zhang, J. Huang, and Z. Li, “Conditional denoising diffusion probabilistic model for ground-roll attenuation,” *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, Nov. 2024.
[16] D. Liu, W. Chen, M. D. Sacchi, and H. Wang, “Should we have labels for deep learning ground roll attenuation?” in *SEG Tech.Program Expanded Abstr.*, Oct. 2020, pp. 3239–3243.
[17] H. Kaur, S. Fomel, and N. Pham, “Seismic ground-roll noise attenuation using deep learning,” *Geophysical Prospecting*, vol. 68, no. 7, pp. 2064–2077, May 2020.
[18] T. Zhong, M. Cheng, X. Dong, Y. Li, and N. Wu, “Seismic random noise suppression by using deep residual U-Net,” *J. Petroleum Sci. Eng.*, vol. 209, p. 109901, Feb. 2022.
[19] N. Pham and W. Li, “Physics-constrained deep learning for ground roll attenuation,” *Geophysics*, vol. 87, no. 1, pp. V15–V27, Nov. 2022.
[20] S. Liu, C. Birnie, A. Bakulin, A. Dawood, I. Silvestrov, and T. Alkhalifah, “A self-supervised scheme for ground roll suppression,” *Geophysical Prospecting*, vol. 72, no. 7, pp. 2580–2598, Apr. 2024.
[21] D. Liu, M. D. Sacchi, X. Wang, and W. Chen, “Unsupervised deep learning for ground roll and scattered noise attenuation,” *IEEE Trans. Geosci. Remote Sens.*, vol. 61, p. 5920613, Oct. 2023.
[22] Y. Li, H. Zhang, J. Huang, and Z. Li, “Conditional denoising diffusion probabilistic model for ground-roll attenuation,” *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, Nov. 2024.
[23] H. Almarzooq and U. bin Waheed, “Automating hyper-parameter optimization in geophysics with optuna: A comparative study,” *Geophysical Prospecting*, vol. 72, no. 5, pp. 1778–1788, Feb. 2024.
[24] Y. Cao, X. Xu, Y. Cheng, C. Sun, Z. Du, L. Gao, and W. Shen, “Personalizing vision-language models with hybrid prompts for zero-shot anomaly detection,” *IEEE Trans. Cybern.*, Feb. 2025.
[25] F. Qian, Y. He, Y. Yue, Y. Zhou, B. Wu, and G. Hu, “Improved low-rank tensor approximation for seismic random plus footprint noise suppression,” *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–19, Feb. 2023.
[26] Y. Chen, M. Bai, and Y. Chen, “Obtaining free usarray data by multi-dimensional seismic reconstruction,” *Nature commun.*, vol. 10, no. 1, p. 4434, Sep. 2019.
[27] Q. Cavalcante and M. J. Porsani, “Low-rank seismic data reconstruction and denoising by cur matrix decompositions,” *Geophysical Prospecting*, vol. 70, no. 2, pp. 362–376, Dec. 2022.
[28] P. Zheng, H. Su, and Q. Du, “Sparse and low-rank constrained tensor factorization for hyperspectral image unmixing,” *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1754–1767, Jan. 2021.
[29] Z. Fang, H. Lin, and X. Xu, “Mask-guided model for seismic data denoising,” *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, Mar. 2022.
[30] D. A. Oliveira, D. G. Semin, and S. Zaytsev, “Self-supervised ground-roll noise attenuation using self-labeling and paired data synthesis,” *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 7147–7159, Oct. 2020.
[31] J. K. Welford and Z. Rongfeng, “Ground-roll suppression from deep crustal seismic reflection data using a wavelet-based approach: A case study from western canada,” pp. 877–884, Jan. 2004.
[32] Y. Li, A. Goodge, F. Liu, and C.-S. Foo, “Promptad: Zero-shot anomaly detection using text prompts,” in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vision*, Apr. 2024, pp. 1093–1102.
[33] P. W. Cary* and C. Zhang, “Ground roll attenuation with

singular value decomposition," in *Global Meeting Abstr.* Society of Exploration Geophysicists, Aug. 2009, pp. 1627–1630.

[34] D. Liu, W. Chen, M. D. Sacchi, and H. Wang, "Should we have labels for deep learning ground roll attenuation?" in *SEG Int. Expo. Annu. Meeting*. SEG, Oct. 2020, p. D031S067R002.

[35] J. Liu, Y. Zhang, J.-N. Chen, J. Xiao, Y. Lu, B. A Landman, Y. Yuan, A. Yuille, Y. Tang, and Z. Zhou, "Clip-driven universal model for organ segmentation and tumor detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vision*, Oct. 2023, pp. 21 152–21 164.

[36] A. Aghayan, P. Jaiswal, and H. R. Siahkoohi, "Seismic denoising using the redundant lifting scheme," *Geophysics*, vol. 81, no. 3, pp. V249–V260, May 2016.

[37] N. Pham and W. Li, "Physics-constrained deep learning for ground roll attenuation," *Geophysics*, vol. 87, no. 1, pp. V15–V27, Nov. 2022.

[38] J. Xia, "Estimation of near-surface shear-wave velocities and quality factors using multichannel analysis of surface-wave methods," *J. Appl. Geophys.*, vol. 103, pp. 140–151, Apr. 2014.

[39] M. U. Khattak, M. F. Naeem, M. Naseer, L. Van Gool, and F. Tombari, "Learning to prompt with text only supervision for vision-language models," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 4, Apr. 2025, pp. 4230–4238.

[40] M. U. Khattak, H. Rasheed, M. Maaz, S. Khan, and F. S. Khan, "Maple: Multi-modal prompt learning," in *Proc. IEEE/CVF Conf. Comput. Vision Pattern Recognit.*, Apr. 2023, pp. 19 113–19 122.

[41] Y. Wu, Y. Zhou, J. Saiyin, B. Wei, and Y. Xu, "Visual textualization for image prompted object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vision*, Jun. 2025, pp. 20 900–20 910.

[42] T. Lüddecke and A. Ecker, "Image segmentation using text and image prompts," in *Proc. IEEE/CVF Conf. Comput. Vision Pattern Recognit.*, Sep. 2022, pp. 7086–7096.

[43] M. Bekara and M. Van der Baan, "Local singular value decomposition for signal enhancement of seismic data," *Geophysics*, vol. 72, no. 2, pp. V59–V65, Feb. 2007.

[44] Y. Chen, Y. Zhou, W. Chen, S. Zu, W. Huang, and D. Zhang, "Empirical low-rank approximation for seismic noise attenuation," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4696–4711, May 2017.

[45] F. Qian, S. Pan, and G. Zhang, *Tensor Computation for Seismic Data Processing: Linking Theory and Practice*. Springer Nature, Apr. 2025, vol. 6.

[46] M. Naghizadeh, "Seismic data interpolation and denoising in the frequency-wavenumber domain," *Geophysics*, vol. 77, no. 2, pp. V71–V80, Feb. 2012.

[47] Y. Chen, S. Jiao, J. Ma, H. Chen, Y. Zhou, and S. Gan, "Ground-roll noise attenuation using a simple and effective approach based on local band-limited orthogonalization," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2316–2320, Sep. 2015.

[48] C. Liang, H. Lin, and H. Ma, "Reinforcement learning-based denoising model for seismic random noise attenuation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, Apr. 2023.

[49] N. Shekhar, D. Tejaswi, A. James, L. Kuruguntla, V. C. Dodda, A. Mandpura, S. Chinnadurai, and K. Elumalai, "Seismic denoising based on dictionary learning with double regularization for random and erratic noise attenuation," *IEEE Trans. Geosci. Remote Sens.*, vol. PP, pp. 1–1, Feb. 2025.

[50] Y. Chen and S. Fomel, "Random noise attenuation using local signal-and-noise orthogonalization," *Geophysics*, vol. 80, no. 6, pp. WD1–WD9, Mar. 2015.

[51] S. Fomel, "Local seismic attributes," *Geophysics*, vol. 72, no. 3, pp. A29–A33, May 2007.

[52] S. A. Hosseini, A. Javaherian, H. Hassani, S. Torabi, and M. Sadri, "Adaptive attenuation of aliased ground roll using the shearlet transform," *J. Appl. Geophysics*, vol. 112, pp. 190–205, Jan. 2015.

[53] V. Kumar, J. Oueity, R. M. Clowes, and F. Herrmann, "Enhancing crustal reflection data through curvelet denoising," *Tectonophysics*, vol. 508, no. 1-4, pp. 106–116, Jul. 2011.

[54] L. Gao, D. Liang, and F. Min, "Unsupervised denoising for seismic data with complementary mask blind spot strategy," *J. Appl. Geophysics*, vol. 221, p. 105307, Feb. 2024.